STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-TENSOR-SYMMETRY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4945-4951 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 4953-4974, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 27 of queue_sdga.json (pb=1021, stmt 4945-4951, proof 4953-4974).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FS5: sources/stacks-sdga/pinned-original-source.tex, source lines 2644–2665. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In the situation above, if $\mathcal{N} \to \mathcal{N}'$ is an isomorphism on cohomology sheaves, then t is an isomorphism of functors $(- \otimes^{\mathbf{L}}_{\mathcal{A}} \mathcal{N}) \to (- \otimes^{\mathbf{L}}_{\mathcal{A}} \mathcal{N}')$.*

Proof. It is enough to show that $\mathcal{P} \otimes_{\mathcal{A}} \mathcal{N} \to \mathcal{P} \otimes_{\mathcal{A}} \mathcal{N}'$ is an isomorphism on cohomology sheaves for any good differential graded $\mathcal{A}$-module $\mathcal{P}$. To do this, let $\mathcal{N}''$ be the cone on the map $\mathcal{N} \to \mathcal{N}'$ as a left differential graded $\mathcal{A}$-module, see Definition 0FS5. (To be sure, $\mathcal{N}''$ is a bimodule too but we don't need this.) By functoriality of the tensor construction (it is a functor of differential graded categories) we see that $\mathcal{P} \otimes_{\mathcal{A}} \mathcal{N}''$ is the cone (as a complex of $\mathcal{O}$-modules) on the map $\mathcal{P} \otimes_{\mathcal{A}} \mathcal{N} \to \mathcal{P} \otimes_{\mathcal{A}} \mathcal{N}'$. Hence it suffices to show that $\mathcal{P} \otimes_{\mathcal{A}} \mathcal{N}''$ is acyclic. This follows from the fact that $\mathcal{P}$ is good and the fact that $\mathcal{N}''$ is acyclic as a cone on a quasi-isomorphism. □

Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.